QUESTION

One test is a multiple choice test with 5 questions, each having 4 options. There are 2000 candidates, each choosing only one answer for each question. Find the smallest possible integer n such that a student's answer sheet, labeled n, has the following property: The answer sheet contains four sheets, and any two of these four sheets have exactly the same three answers.

Teacher stronger — Student stronger

RESPONSE

Hmm, I need four answer sheets such that any two have exactly three entries the same, so another way to think about this is that for any pair of sheets, 3 out of 5 entries are the same, and 2 are different. This seems to be a very restrictive condition.

Wait, maybe I can think of it in terms of combinations. When you have two sheets which have three answers the same, how many possible combinations are there for that pair? If we have three questions matching and two different, then there are 5 choose 3 (the number of questions where the answers match), which is 10. But then once you pick three positions, the answers on those positions can be anything, but the other positions need to be different.

However, I think that to pair any two sheets, for them to share three answers, they need to also have their unique answers on the other two question positions. So that might require some careful consideration.

There is a problem because I have 4 sheets, so there are 4 choose 2 = 6 pairs of sheets, and each of these pairs must have three answers the same.

Wait, maybe I can first think of a smaller example, and then maybe generalize.

Wait, let me see. Let me try to think of a smaller example especially. Suppose I have two answer sheets. The requirement would be that they share exactly 3 correct answers. That would mean, if the first sheet is [AB CD]. Then the second sheet has to have 3 of the same answers as the first one.

But here, I can take two different sheets, for instance, [0 0 0 0 0] and [0 0 0 0 1], which have 4 answers the same. So in that case, it does not satisfy the condition. So maybe my first example fails.

Wait, so if I have two sheets, they cannot have all four choices the same. They all must share exactly three same answers. Let me think of an example.

Let me see. For two sheets to have three answers the same, and two different, if I have A=[0 0 0 0 0] and B has three 0's and two different. Suppose B=[0 0 0 1 1]. That would work.